

Special topics in logic and security II *

May 14, 2022

1. A code C consists of the following four words:

$$(0, 0, 0, 0, 0, 0, 0, 0); (1, 1, 1, 1, 0, 0, 0, 0); (0, 0, 0, 0, 1, 1, 1, 1); (1, 1, 1, 1, 1, 1, 1, 1).$$

- (a) Show that this code is linear over $\mathbb{F}_2$. Find its parameters $[n, k, d]$.
- (b) Find a generating matrix G for C .
- (c) Find a check-matrix H for C .
- (d) Show that $C \subset C^\perp$.
- (e) Find the minimal distance of $C^\perp$.

2. Let C be the following code over the field $\mathbb{F}_{11}$.

$$C = \{(f(1), f(2), \dots, f(7)) \mid f(x) = ax^2 + bx + c \in \mathbb{F}_{11}[x]\}.$$

- (a) Show that one can always correct one mistake.
- (b) Correct the word $(3, 7, 2, 10, 9, 10, 8)$.

3. Suppose that $x \in H_n$ and $y \in H_m$ are quantum states. Show that $x \otimes y \in H_n \otimes H_m$ is a quantum state.

4. On a table at CERN there are two independent globes. The red globe has the quantum state $r = \frac{1-i}{2}|0\rangle + \frac{1+i}{2}|1\rangle$ while the blue globe has the quantum state $b = \frac{1+i}{2}|0\rangle + \frac{1-i}{2}|1\rangle$. We interpret by $|1\rangle$ that a mutation happens in the globe.

- (a) How probable is to separately observe mutations in these globes?
- (b) We consider the composite system of the two globes. Compute its quantum state.
- (c) How probable is to observe a mutation in the composite system?

5. Let H_n be a complex Hilbert space. The state $|a_1\rangle \in H_n$ is used as a blank symbol. A unitary function $U : H_n \otimes H_n \rightarrow H_n \otimes H_n$ such that for every state $|x\rangle \in H_n$ one has:

$$U(|x\rangle|a_1\rangle) = |a_1\rangle|a_1\rangle$$

is called quantum delete machine. Show that there is no quantum delete machine.

6. OPTIONAL QUESTION

A triangle ABC has angles A, B and C. Show that the triangle is acute if and only if there is another triangle UVW with side-lengths A, B and C.

*Master, 4-th semester.